AKSHAY CHANDAR M

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Professional Summary

Systems-oriented Software Engineering student focused on object-oriented programming, distributed data architectures, and high-performance backend pipelines. Proven track record handling massive time-series telemetry streams (10M+ records), engineering low-latency data pipelines, and implementing secure automation layers. Backed by advanced algorithmic problem-solving abilities and a growth mindset dedicated to optimizing software availability, efficiency, and reliability at scale.

Education

Kongu Engineering College, Erode, Tamil Nadu, India *2024 – 2028*
B.E. Computer Science and Design **CGPA: 8.40 / 10.0**
Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems.

Technical Skills

Languages & Databases:	Python · Java · SQL · Redis · PostgreSQL · MySQL · MongoDB
Backend & Distributed Systems:	REST APIs · Microservices · System Design · Node.js · Express.js · FastAPI
Data Engineering & Applied ML:	CatBoost · Apache Parquet · Time-Series Modeling · Feature Engineering
DevOps, Cloud & Observability:	Linux · Docker · Kubernetes · Jenkins (CI/CD) · Prometheus · Git

Projects

SolarSentinel-Prime: Real-Time Satellite Telemetry Pipeline *Python, Apache Parquet, CatBoost, Streamlit, Linux Systems*

Bharatiya Antariksh Hackathon 2026 — Indian Space Research Organisation (ISRO) *Jun 2026 – Present*

- Developed an automated, real-time solar flare predictive pipeline processing 10.1M+ rows of flight data from the SoLEXS and HEL1OS payloads on the Aditya-L1 satellite, boosting ingestion throughput by 10x via native Linux filesystem migration.
- Optimized system memory footprint by 85% by designing an out-of-core precompiler script that dynamically streams, cleans, and serializes massive multi-instrument datasets day-by-day into snappy-compressed Apache Parquet binary blocks.
- Eliminated active-region spatial data leakage and target proxy bias by structuring a strict chronological validation split (75% train/25% test), raising model target recall from 21.34% to 80.78% via relative, scale-invariant physics features.
- Trained a cost-sensitive CatBoost classification framework with asymmetric loss weighting (1.0 to 2.5), securing a 74.24% Operational Precision and 20-minute advanced warning lead times before peak impulse spikes.
- Built a sub-millisecond low-latency inference delivery framework with an interactive Streamlit dashboard, enabling operators to scrub through streaming multi-channel light curves with automated visual alert triggers.

Distributed High-Concurrency Scraper *Java, Python, Redis, Multi-threading, Scrapy*

- Engineered a high-performance distributed data-ingestion engine that accelerated throughput by 40% by managing concurrent network operations across asynchronous, multi-threaded Java thread pools.
- Reduced external request latency and redundant resource overhead by implementing an atomic, synchronized caching abstraction tier using a centralized Redis memory layer.
- Prevented system memory depletion under high traffic loads by structuring a decoupled, high-availability ingestion queue driven by greedy priority sorting algorithms to manage payload spikes safely.

Git-it-Done: Automated GitHub Issue Solver *Python, GitHub API, Docker, LLM Engineering*

- Automated end-to-end software defect remediation workflows by designing an event-driven system built on contextual file-tree parsing and LLM orchestration APIs to generate deterministic codebase patches.
- Eliminated code regression and security injection risks by architecting a programmatic testing harness that executes and validates generated patches inside short-lived, isolated Docker runtime sandboxes.
- Streamlined developer code review cycles by integrating directly with the secure GitHub REST API tier to automate patch payload formatting and Pull Request proposals based on specified feature requirements.

AnaMetrix: DevOps Metrics Dashboard *Python, FastAPI, Prometheus, MongoDB, Docker, React*

- Standardized infrastructure observability and product quality tracking by engineering responsive Python/FastAPI microservices to monitor core DORA metrics, pipeline performance, and deployment frequency.
- Optimized telemetry data ingestion paths by mapping unstructured cluster operational events into structured target database documents using optimized Prometheus time-series queries.
- Achieved 100% deployment environment parity and eliminated local system discrepancies by structuring clean service isolation via standardized Docker Compose network topologies.

Technical Achievements

- **Data Structures & Algorithms:** Solved **670+** competitive problems across **LeetCode** (54% Medium/Hard ratio) and GeeksforGeeks; achieved a **1,554 peak contest rating** (Top 31.39% globally) demonstrating strong performance under timed constraints.
- **Smart India Hackathon 2025:** Selected as a national Semi-finalist among a pool of 5,000+ competitive teams; served as Technical Lead directing cross-functional collaboration, backend routing architectures, and system logic.
- **Continuous Infrastructure Practice:** Maintained active system tiering status within the KodeKloud Engineer matrix, completing real-time practical evaluations across Linux internal subsystems and cluster management setups.